Sangjin Choi

Postdoctoral Researcher
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RESEARCH STATEMENT

I design systems software for cost-efficient AI infrastructure by exploiting hardware and workload characteristics.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Ph.D. in School of Computing — Sep 2021 – Aug 2026
Advisor: Youngjin Kwon, CASYS Lab

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Master of Science in School of Computing — Sep 2019 – Aug 2021
Advisor: Youngjin Kwon, CASYS Lab

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Bachelor of Science in School of Computing — Mar 2015 – Aug 2019

WORK EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Postdoctoral Researcher, Information & Electronics Research Institute — Sep 2026 – Present

Microsoft Research Asia — Vancouver, Canada
Research Intern, Vancouver Lab — Mar 2026 – May 2026

Microsoft Research Asia — Beijing, China
Research Intern, Systems and Networking Research Group — Jul 2025 – Jan 2026

PUBLICATIONS

[1] [**ACM ATC 2026**] **Sangjin Choi**, Sukmin Cho, Yifan Xiong, Ziyue Yang, Youngjin Kwon, and Peng Cheng. "ELDR: Expert-Locality-Aware Decode Routing for PD-Disaggregated MoE Serving."

[2] [**MLSys 2026**] Hyunjae Lee, **Sangjin Choi**, Seungjae Lim, and Youngjin Kwon. "BEAM: Joint Resource–Power Optimization for Energy-Efficient LLM Inference under SLO Constraints."

[3] [**ACM EuroSys 2026**] Changjun Lee, **Sangjin Choi**, and Youngjin Kwon. "MTTM: Dynamic Fast Memory Partitioning with Bandwidth Optimization for Multi-tenant Cloud."

[4] [**Findings of NAACL 2025**] Sukmin Cho, **Sangjin Choi**, Taeho Hwang, Jeongyeon Seo, Soyeong Jeong, Huije Lee, Hoyun Song, Jong C. Park, and Youngjin Kwon. "Lossless Acceleration of Large Language Models with Hierarchical Drafting based on Temporal Locality in Speculative Decoding."

[5] [**IEEE/ACM ICCAD 2023**] Jaehoon Heo, Yongwon Shin, **Sangjin Choi**, Sungwoong Yune, Jung-Hoon Kim, Hyojin Sung, Youngjin Kwon, and Joo-Young Kim. "PRIMO: A Full-Stack Processing-in-DRAM Emulation Framework for Machine Learning Workloads."

[6] [**USENIX ATC 2023**] **Sangjin Choi**, Inhoe Koo, Jeongseob Ahn, Myeongjae Jeon, and Youngjin Kwon. "EnvPipe: Performance-preserving DNN Training Framework for Saving Energy."

[7] [**USENIX ATC 2022**] **Sangjin Choi**[*], Taeksoo Kim[*], Jinwoo Jeong, Rachata Ausavarungnirun, Myeongjae Jeon, Youngjin Kwon, and Jeongseob Ahn. "Memory Harvesting in Multi-GPU Systems with Hierarchical Unified Virtual Memory." ([*]Equal contribution.)